Introduction to Pharmacology



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Pharmacology: The study of substances that interact with living systems through biochemical processes.

Its derived from Greek words: Pharmacon = Drug

logos = science or study

Clinical Pharmacology: Is the science of substances used to prevent, diagnose and treat disease.

Toxicology: Is that branch of pharmacology that deals with the undesirable effects of chemical agents on living systems.

Food and Drug Administration (FDA): The federal agency responsible for the safety and efficacy of all drugs in the United States, as well as food and cosmetics

Drug: a drug may be defined as any substance that produce a change in biologic function through its chemical actions.

DRUG is French word: drogue means dried herbs

Drug nomenclature

1-Chemical name

7-chloro-1-methyl-5-phenyl-1,3-dihydro-2H-1,4-

benzodiazepine-2-one.

2-Generic name:

An official generic name is given to a drug by national and international expert bodies—such as the United States Adopted Names (USAN) Council working together with the World Health Organization (WHO)—using a standardized naming system based on the drug's chemical class and mechanism of action

3-Brand name / Trade name:

It is the name given to a drug by the company which manufactures drug. Each company has thier own brand name.

A drug may be prescribed either by its generic name / official name or by Brand name / Trade name.

In Pharmacology, only the generic name should be used, not the brand names.

Generic name	Brand name
(1) Paracetamol	Tylenol
(2) Diazepam	Valium
(3) Metronidazole	Flagyl

Nature and sources of drugs

1-Plant source

Aspirin, Paclitaxel, Morphine, Digoxin

2-Animal source

Heparin, insulin, L. Thyroxine

3-Microorganism

Penicillin, Polymyxin B

4-Mineral sources:

Magnesium sulfate, Ferrous sulphate

5-Synthetic drugs

Metronidazole, more than 90% of the drugs are synthetic in nature (manufactured in the drug industry)

6-Recombinant DNA technology Recombinant

Human Insulin, growth hormone and interferons

In this methode a desired gene is inserted into a very fast multiplying non-pathogenic strain of some bacteria e.g., E.coli-K12. This host cell will now produce large amount of the gene-directed proteins (DRUGS), which are required. E.coli, otherwise, does not synthesize these proteins.

Example: Humulin (a human Insulin synthesized by inserting proinsulin gene). Humulin is very closely resembles human Insulin.

Some Terms Related to Drugs

Prototype drug: is a well-studied drug that serves as the standard model and reference point for a whole class or group of similar drugs.

Example: **Atropine** is aprtotype of Anticholenergic drugs

Prodrug: It is an inactive drug which become active when metabolized in the body.

Example:Codeine: Is broken down by the liver into morphine

Onset of action of a drug: The length of time it takes for a medicine to start to work.

Duration of action of adrug: The length of time that a particular drug is effective in the body.

Side effects : A side effect is any unintended effect of a drug that occur at doses normally used by a patient which is related to the pharmacological properties of the drug.

Example:Side effects of Metformin:

Nausea, flatulence and rarely lactic acidosis and Wight loss

A drug idiosyncrasy: is an abnormal or unexpected reaction to a drug that occurs in a small number of people and has no relation to the dose taken.

Example: Carbamazepine: Stevens-Johnson syndrome:

Heparin: Thrombocytopenia

Chloramphenicol: Aplastic anemia

Pharmacokinetics: It means the movement of the drug within the body it includes the processes of absorption (A), distribution (D), metabolism (M) and excretion (E). It means ‘what the body does to the drug’.

Pharmacodynamics: It is the study of drugs—their mechanism of action, pharmacological actions and their adverse effects. It covers all the aspects relating to ‘what the drug does to the body’.

Pharmacogenetics: is defined as the study of variability in drug response due to heredity.

Pharmacogenomics: This recent term overlaps with pharmacogenetics, describing the use of genetic information to guide the choice of drug therapy on an individual basis.

Patients with breast cancer are tested for the **HER2 gene:** If the tumor overexpresses this gene, targeted therapies like **Herceptin** work effectively

Pharmacovigilance: It is the science related to the Detection, Assessment, Understanding and Prevention (DAUP) of adverse effects or any other drug related problems.So, it means continuous monitoring for unwanted effects of marketed drugs

References

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